Report

Of

E-COMMERCE WEBSITE WITH MODERN UI

(MERN Full Stack Development With Project — Virtual Internship)

Submitted To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech CSE

By

Name of Student: BHASKAR SINGH
Roll Number of Student: CS-23411470
Batch: 2026-27

CANDIDATE'S DECLARATION

I, Bhaskar Singh, do hereby declare that the internship report titled “E-Commerce Website with Modern UI” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own work and has not been submitted to any other institute for any degree/diploma requirement, and I shall be solely responsible for any kind of copyright violation in this regard.

Signature
Student Name: Bhaskar Singh
Roll No.: CS-23411470

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the MERN Full Stack Development With Project Virtual Internship. I am thankful for their aspiring guidance, invaluably constructive criticism, and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this project.

I would like to thank the School of Computer Science and Engineering, IILM University, Greater Noida, for providing me the opportunity to undertake this virtual internship, and EduSkills Academy, along with AICTE's National Internship Portal, for organising and mentoring the program.

I express my gratitude to my parents for their blessings and constant encouragement.

Signature
Student Name: Bhaskar Singh
Roll No.: CS-23411470

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IILM UNIVERSITY

INTERNSHIP COMPLETION CERTIFICATE

शिक्षा मंत्रालय
MINISTRY OF EDUCATION

NATIONAL INTERNSHIP PORTAL

EduSkills
Nation Building Through Skills

AICTE - EduSkills 2026 VIRTUAL INTERNSHIP

Certificate of Virtual Internship

This is to certify that

Bhaskar Singh

IILM University, Greater Noida

has successfully completed the 8-weeks

MERN full Stack Development With Project Virtual Internship

During June - August 2026

Supported by EduSkills ACADEMY

Dr. Buddha Chandrasekhar
Chief Coordinating Officer (CCO)
AICTE, Ministry of Education

Shubhajit Jagadev
Chief Executive Officer (CEO)
EduSkills

Certificate ID : 4512a80dca9753aef3bc

Student ID: STU67e683116c83d1743160081

INTERNSHIP GRADE O

GRADE- O (Outstanding):90-100 | E (Excellent):80-89 | A (Very Good):70-79 | B (Good): 60-69 | C (Fair): 50-59 | D (Average): 40-49 | P (Pass): 30-39 | F (Fail): Below 30

4. PROJECT DESCRIPTION

4.1 Introduction

The MERN Full Stack Development With Project Virtual Internship is an 8-week AICTE-approved program delivered through the National Internship Portal in association with EduSkills Academy. The internship is designed to give students hands-on, industry-relevant exposure to full stack web development using the MERN stack — MongoDB, Express.js, React.js, and Node.js — which together form one of the most widely adopted technology stacks for building modern, scalable, and responsive web applications.

As part of this internship, I undertook the design and development of an E-Commerce Website with a modern user interface. The project was chosen to apply the concepts learned during the internship — component-based front-end development, RESTful API design, NoSQL database modelling, authentication, and state management — to a real-world use case that online businesses rely on every day. This chapter documents the organization behind the program, the problem the project addresses, its objectives and scope, the technologies used, the system architecture, the methodology followed, and the outcomes achieved.

4.2 Organization Profile

This virtual internship was conducted under the AICTE (All India Council for Technical Education) National Internship Portal, an initiative of the Ministry of Education, Government of India, aimed at providing structured, skill-based internship opportunities to technical students across the country. The training and mentorship for the program were delivered by EduSkills Academy, an ed-tech organization that partners with AICTE to design and run industry-aligned virtual internship tracks in emerging technology domains such as full stack development, cloud computing, data science, and cybersecurity.

Through structured weekly modules, recorded and live sessions, hands-on assignments, and a capstone project, EduSkills Academy enables students to build practical, portfolio-ready skills that are directly relevant to the software industry, while AICTE certification lends the program formal academic recognition.

4.3 Problem Statement

Small and medium businesses that wish to sell products online are often limited by expensive, rigid, or poorly designed e-commerce platforms. Many existing solutions either require significant licensing costs, offer outdated and non-responsive user interfaces, or are difficult to customize for a specific brand's needs. Customers, in turn, expect a fast, intuitive, and mobile-friendly shopping experience with smooth product discovery, a reliable cart and checkout flow, and clear order tracking.

The problem addressed by this project was to design and build a full-featured, open, and cost-effective e-commerce web application with a modern, clean, and responsive user interface — one that supports product browsing and search, cart and wishlist management, secure user authentication, order placement, and an admin panel for managing products and orders — using a scalable full stack JavaScript architecture.

4.4 Project Objectives

- To design a responsive, modern, and user-friendly e-commerce interface using React.js.
- To build a secure and scalable back-end using Node.js and Express.js, exposing RESTful APIs.
- To design a flexible NoSQL data model in MongoDB for products, users, carts, and orders.
- To implement user authentication and authorization using JSON Web Tokens (JWT).
- To build core shopping features: product listing, search and filters, product details, cart, wishlist, and checkout.
- To provide an admin dashboard for managing products, categories, orders, and users.
- To apply industry-standard practices such as version control, modular code structure, and API testing.

4.5 Scope of the Project

The scope of the project covers the design and development of a single-vendor e-commerce web application, including customer-facing features (browsing, search, filtering, cart, wishlist, checkout, and order history) and an administrative interface (product, category, order, and user management). The application supports secure user registration and login, role-based access for admin and customer accounts, and integration points for payment processing.

The project does not, within the internship timeline, extend to multi-vendor marketplace functionality, live third-party payment gateway certification, or production-scale deployment with load balancing — these are identified as areas for future enhancement.

4.6 Technologies and Tools Used

Category	**Technology / Tool**
Front-end Library	React.js (with React Router, Hooks, Context API)
Styling	Tailwind CSS for a modern, responsive UI
State Management	Redux Toolkit
Back-end Runtime & Framework	Node.js, Express.js
Database	MongoDB with Mongoose ODM
Authentication	JSON Web Tokens (JWT), bcrypt for password hashing
Payment Integration	Razorpay / Stripe test integration
Image/Media Storage	Cloudinary
API Testing	Postman
Version Control	Git and GitHub
Development Tools	Visual Studio Code, Chrome DevTools

4.7 System Architecture

The application follows a three-tier client-server architecture, a natural fit for the MERN stack, cleanly separating presentation, business logic, and data storage.

Layer	**Responsibility**	**Technology**
Presentation (Client) Layer	Renders the UI, handles user interaction and client-side routing	React.js, Tailwind CSS, Redux Toolkit
Application (Server) Layer	Exposes REST APIs, implements business logic, authentication, and validation	Node.js, Express.js, JWT
Data (Database) Layer	Stores and retrieves persistent data — users, products, carts, orders	MongoDB, Mongoose

The React.js front end communicates with the Express/Node.js back end over HTTPS using JSON-based RESTful APIs. The back end validates requests, applies business logic, authenticates users via JWT, and interacts with MongoDB through the Mongoose ODM (Object Data Modelling) library. This layered design keeps the codebase modular, makes independent scaling of the front end and back end possible, and allows each layer to be developed, tested, and deployed independently.

4.8 Methodology

The project was developed iteratively over the 8-week internship duration, following an Agile-inspired, sprint-based methodology. Each week combined structured learning with incremental, hands-on implementation of the project.

Week	Focus Area
Week 1–2	Orientation, MERN stack fundamentals, and project planning; requirement analysis and database schema design
Week 3–4	Back-end development — Express.js REST API, MongoDB models, JWT authentication
Week 5–6	Front-end development — React component design, routing, state management, and UI integration with APIs
Week 7	Cart, checkout, payment integration, and admin dashboard development
Week 8	Testing, bug fixing, UI polishing, deployment preparation, and final project presentation

4.9 Expected Outcomes

By the end of the internship, the project achieved the following outcomes:

- A fully functional, responsive e-commerce website with a modern UI, tested across desktop and mobile screen sizes.
- Secure user registration, login, and JWT-based session handling.
- Product catalogue with search, category filters, and detailed product pages.
- Shopping cart, wishlist, and a multi-step checkout flow with order summary.
- An admin dashboard to add, update, and remove products and manage orders.
- RESTful API layer tested using Postman, with clear separation of routes, controllers, and models.

- Practical, hands-on experience with the complete MERN development workflow, from database design to deployment-ready builds.

4.10 Certificates of Completion and Other Communication/Mail Proof

A copy of the AICTE – EduSkills Certificate of Virtual Internship, awarded on successful completion of the 8-week MERN Full Stack Development With Project Virtual Internship with an "Outstanding" grade, is attached in Chapter 3 of this report (Certificate ID: 4512a80dca9753aef3bc, Student ID: STU67e683116c83d1743160081). Copies of related program communication and completion e-mails may be attached here as additional appendices, as applicable.

5. BIBLIOGRAPHY / REFERENCES

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